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Project Reflection

For our project, we worked on building an automated awning system for restaurant workers. It allows them to focus more on what really matters, giving their customers exceptional service and ensuring the awning for outdoor customers would be deployed at the right times. Let's say it was a sunny day, the awning would not be deployed and the customers could enjoy the beautiful weather. But let's say the weather wasn't as beautiful, maybe it was really humid or wet out, the awning would automatically deploy so the workers don't have to worry about complaints from the customers of their food getting wet from rain.

To start our project we began planning, asking big questions like what materials will we use to make this or what will our circuitry diagram look like? As a group we created a flow chart to make sure we were all on the same page of what we were designing. Then it was time to get working, because none of us had much experience really anywhere, we thought it would be best to take things little by little. During this project, I was responsible for most of the circuitry/electronics, Will was responsible for most of the code, and Brent led the mechanical design. I started out by using a button and a LED, the LED would indicate whether or not the system was on or not. The button would be a switch that turned the system on or off. I wired the LED to be connected to an analog pin in our arduino and put the button to be connected to a digital pin in our arduino. Will put something like whenever the button is pressed, I want the LED to turn on and stay on or turn off and stay off until the button is pressed again as the prompt.

The next major step was using our motor that would function when the system was on. I wired the motor to another digital pin. The idea we had to give to ChatGPT to create the correct code was when there was voltage going through our analog pin, the system was on, therefore the motor should spin. It took us some time to figure out how to get ChatGPT to create the code we wanted, but eventually we got it. The final step was incorporating our custom made sensor, using alligator clips and jumper wires I wired our custom sensor to another digital pin. This time for the code, Will had to make sure ChatGPT would allow for the motor to rotate the awning whenever the system was on and our custom sensor was sensing moisture (our breaths). We also had to find the right amount of rotation for the servo motor so our awning wouldn't overrotate or underrotate. This was about all the code and electronics we had to do.

Brent was the one mostly responsible for creating the structure. For our structure, we ended up using popsicle sticks, tape, and hot glue. The awning itself was made out of cardboard and so were the walls of our restaurant. We also built a little motor mount out of popsicle sticks to ensure the motor was held in place. Once we had our structure, we had put everything together to make our final product!